THURSDAY DEMONSTRATION

Search in Large Spaces

Without a Utility Function

From natural-language game ideas to executable rules and playable experiments.



DEMO OBJECTIVE

Generate → validate → play → evaluate

A small independent Ludii example is working; the team pipeline is under development.

Proposed LLM → Ludii pipeline

1

Prompt + examples

Describe a new game

→

2

Generate + critique

LLM revision loop

→

3

Validate + filter

Syntax, legality, termination

→

4

Save + load

Ludii runner / bridge

→

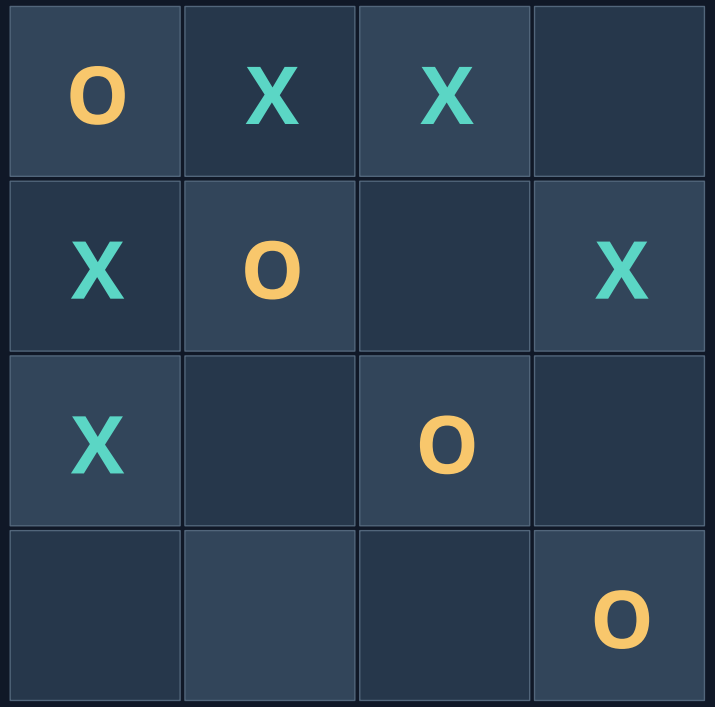
5

Play + evaluate

AI matches and game metrics

STATUS: Pipeline proposed; independent game creation and AI-vs-AI play demonstrated.

A working Ludii game: Four-in-a-Row



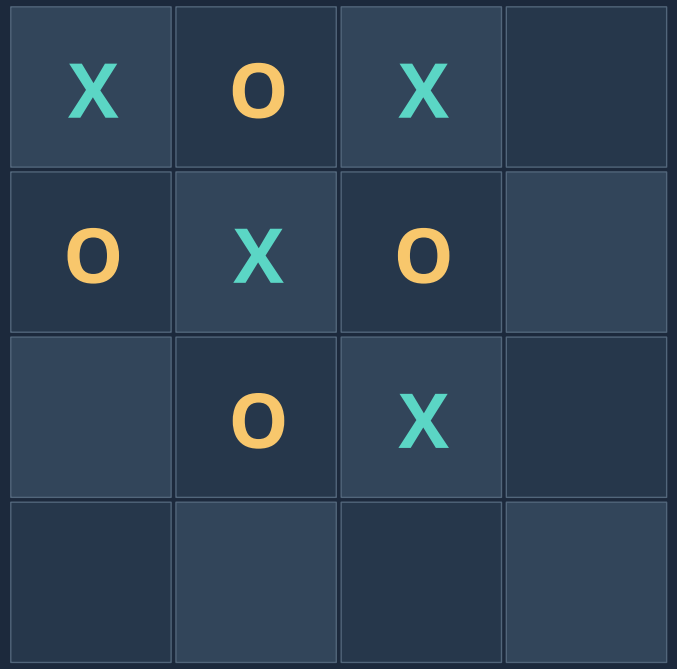
Rules encoded in a compact .lud file

```
(game "Four-in-a-Row"
  (players 2)
  (equipment {
    (board (square 4))
    (piece "Disc" P1)
    (piece "Cross" P2) })
  (rules
    (play (move Add (to (sites Empty))))
    (end {
      (if (is Line 4) (result Mover Win))
      (if (is Full) (result Mover Draw)) })))
```

Compiled • legal moves tested • victory displayed • trial saved

LIVE DEMONSTRATION

AI agents can play the generated game



Reproducible workflow

1. Load the .lud game
2. Select two AI players
3. Restart → Play/Pause
4. Observe result and save .trl

Goal: working AI-vs-AI play, not high chess-playing strength.

Three games, radically different complexity

3×3 Tic-Tac-Toe

9 squares • 8 winning lines

X	O	X
O	X	O
O	X	O

Optimal result: draw

Elementary analysis; ancient game

4×4 Four-in-a-Row

16 squares • 10 winning lines

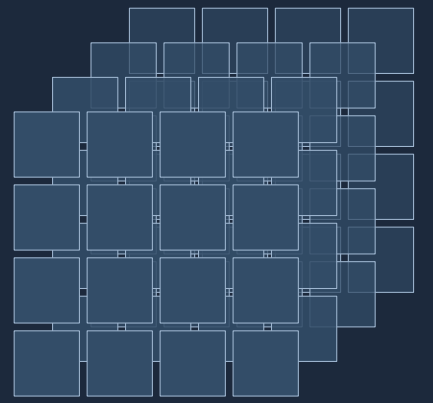
X	O		
O	X		
		O	
			X

Optimal result: draw

Non-search proof: J. Uiterwijk (2017/18)

4×4×4 Qubic

64 cells • 76 winning lines



First player can force a win

O. Patashnik (1980); K. Thompson verified

State-space upper bounds: $3^9 \approx 2\times10^4$ • $3^{16} \approx 4\times10^7$ • $3^{64} \approx 3\times10^{30}$

Qubic: ~1,500 computer hours (1970s); Thompson independently checked Patashnik's proof (~50 hours).

Status, research questions, next steps

WORKING NOW

- ✓ 4×4 game compiles and loads
- ✓ Legal moves and winning condition tested
- ✓ AI vs AI played; trial saved
- ✓ Standalone demo available

NEXT RESEARCH STEPS

- → Integrate teammate-generated games
- → Automate LLM critique and revision
- → Validate and filter candidate games
- → Study game metrics and skill growth

Research question: Can measurable game properties predict enduring strategic interest?